

Scanner Buffer issue in Java

COP 3330

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```
System.out.println("Enter age");
int age;
age = scan.nextInt();
System.out.println("Age is "+age);
System.out.println("Enter full name");
String name = scan.nextLine();
System.out.println("Name is <"+name+">");
System.out.println("Enter name of school");
String school = scan.nextLine();
```

Enter age

5

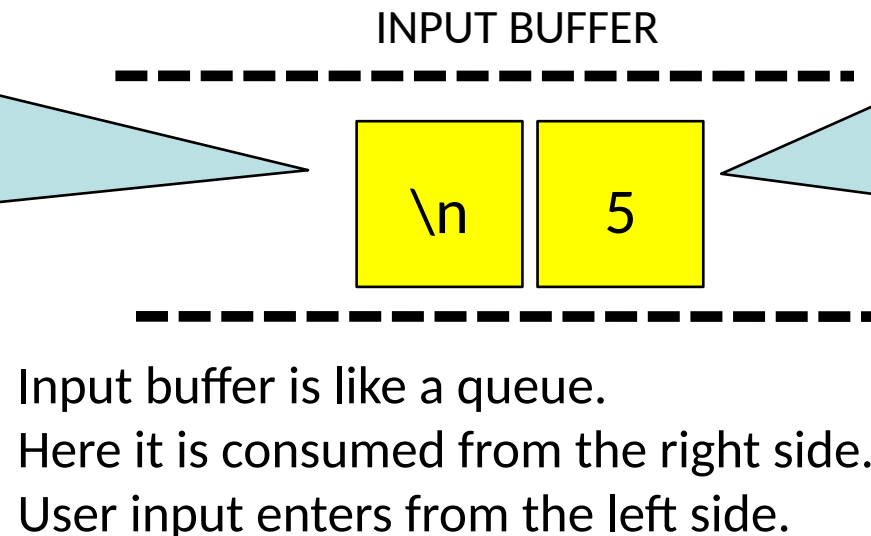
Age is 5

Enter full name

Name is<>

Enter name of school

As `nextLine()`, I read everything in the buffer until I see a newline (enter character). Apart from `\n`, I read nothing legally. It was ending with newline.



As `nextInt()`, I reads characters up to — **but not including** — the next empty space character (tab, space, enter), and try to parse that chunk as an int.

Full code of the problem case

```
package testjava;
import java.util.Scanner;
public class test{
    public static void main(String[] args){
        Scanner scan=new Scanner(System.in);
        System.out.println("Enter age");
        int age;
        age = scan.nextInt();
        System.out.println("Age is"+age);
        System.out.println("Enter full name");
        String name = scan.nextLine();
        System.out.println("Name is <"+name+">");
        System.out.println("Enter name of school");
        String school = scan.nextLine();
    }
}
```

**The test.java is
inside testjava
package.
Scanner is imported.**

What is the problem?

- `scan.nextInt()` only reads until new line/tab/space character
- If this new line character stays in the buffer
 - It creates problem for other read operations
 - The other read operation will read what is “left” in the buffer.
 - Someone needs to clean the buffer when we will read from it.
 - NO! There is no `buffer clear` in Java. The only appropriate way to read character from the buffer so that it will be removed from the buffer.

How to fix it?

- **The only way to remove a character from buffer is to read it.**
 - You may read and not use that character
- We have 2 solutions for this
 - Do NOT use next.., Always use nextLine(). If you need to read only one word, parse the string for the first word. Ignore the remaining if the user has input more than one word
 - Consume the newline character whenever you use next(),nextInt() etc.. since it is not consumed automatically.All the nextXXXX() and next() functions read until empty space. They won't read empty space.

```
System.out.println("Enter age");
int age;
age = scan.nextInt();
scan.nextLine();
System.out.println("Age is"+age);
System.out.println("Enter full name");
String name = scan.nextLine();
System.out.println("Name is <"+name+">");
System.out.println("Enter name of school");
String school = scan.nextLine();
```

Enter age

5

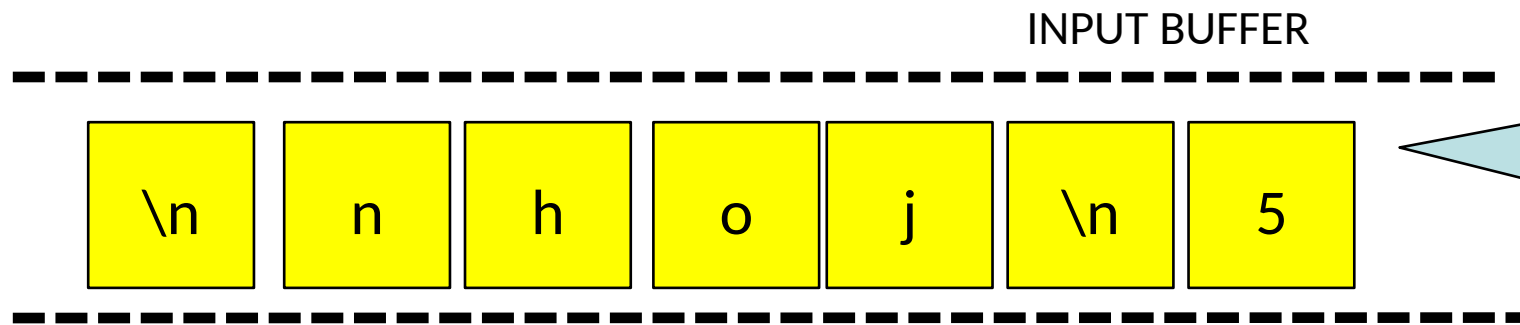
Age is 5

Enter full name

john

Name is <john>

Enter name of school



Input buffer is like a queue.
Here it is consumed from the right side.
User enters input from the left side.

Enter character(`\n`) after 5 is removed(read) by the `scan.nextInt();` line. It is read from buffer but ignored afterwards.

Fixed code of the problem case

```
package testjava;
import java.util.Scanner;
public class test{
    public static void main(String[] args){
        Scanner scan=new Scanner(System.in);
        System.out.println("Enter age");
        int age;
        age = scan.nextInt();
        scan.nextLine();
        System.out.println("Age is"+age);
        System.out.println("Enter full name");
        String name = scan.nextLine();
        System.out.println("Name is <"+name+">");
        System.out.println("Enter name of school");
        String school = scan.nextLine();
    }
}
```

**The test.java is
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Another solution

- Do not use `nextInt()` for our case.
- Read the user input with `nextLine()`
- Then convert the `String` input(`age`) to `Integer`
- Please try to code before seeing the solution

Fixed code of the problem case

```
package testjava;
import java.util.*;
public class test{
    public static void main(String[] args){
        Scanner scan=new Scanner(System.in);
        System.out.println("Enter age");
        int age;
        String acquired=scan.nextLine(); //this is our age in string
        age = Integer.parseInt(acquired); //convert it to int
        System.out.println("Age is "+age);
        System.out.println("Enter full name");
        String name = scan.nextLine();
        System.out.println("Name is <"+name+">");
        System.out.println("Enter name of school");
        String school = scan.nextLine();
    }
}
```

Some scenarios to try

- Try inputting a space before entering age, name etc.. and see what happens
- Try inputting space after entering age, name and see what happens.
- Try asking user their salary as float after age.
 - Then fix the code (or write the unfixed(broken) code first)
- Try those with the fix and without the fix.